

Modeling 2D Marshak Waves

Understanding the physics of x-ray radiation
driven supersonic nonlinear heat waves

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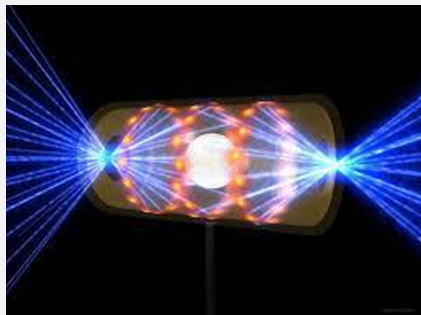
Outline

- 1 Describing the Physical system
- 2 Marshak Waves Experimental Setup
- 3 1.5D model
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 - 2D effects
- 4 Full 2D model
 - Modeling the front curvature
 - Comparing to 2D supersonic diffusion simulation
 - A full 2D Temperature distribution
 - Comparison with Experimental Data
- 5 Conclusion

Motivation

Radiative heat waves are highly common in High Energy Density Physics:

Inertial Confinement Fusion



- Driven by radiation in Hohlraums (e.g., at NIF).

Astrophysics



- Crucial for energy transport in stars.

The Governing Equations

Scattering-free gray Boltzmann equation for photons (RTE):

$$\frac{1}{c} \frac{\partial I(\hat{\Omega}, \vec{r}, t)}{\partial t} + \hat{\Omega} \cdot \vec{\nabla} I(\hat{\Omega}, \vec{r}, t) = \sigma'_a(T_m) \left(B(T_m(\vec{r}, t)) - I(\hat{\Omega}, \vec{r}, t) \right) \quad (1)$$

Energy exchange between radiation and matter:

$$\frac{C_v(T_m)}{c} \frac{\partial T_m}{\partial t} = \sigma'_a(T_m) \left(\frac{1}{c} \int_{4\pi} I d\hat{\Omega} - a T_m^4 \right) \quad (2)$$

Gray Coupled Equations and LTE

Integrating the previous two equations, and using the diffusion approximation, we get the following coupled equations,

$$\begin{aligned}\frac{\partial E}{\partial t} - \nabla \cdot \left(\frac{c}{3\sigma} \nabla E \right) &= c\sigma(aT_m^4 - E), & E &= aT_R^4 \\ \rho \frac{\partial e}{\partial t} &= c\sigma(E - aT_m^4)\end{aligned}$$

Finally, assuming thermodynamic equilibrium (TE) between the radiation field and the material ($T \equiv T_R = T_m$):

$$\frac{\partial T^4}{\partial t} + \rho \frac{\partial e}{\partial t} = \nabla \cdot \left(\frac{c}{3\sigma} \nabla (aT^4) \right) \quad (3)$$

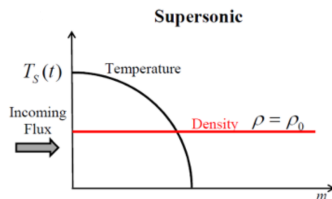
The Problem

- Power-law approximation for specific energy and opacity:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\sigma} = \frac{1}{\rho\kappa} = gT^\alpha \rho^{-\lambda}$$

The Core Challenge

The diffusion coefficient $D = \frac{c}{3\rho\kappa}$ and the material internal energy e are highly non-linear. The system becomes very difficult to solve analytically.



Shussman et al. POP, (2015)

The Goal

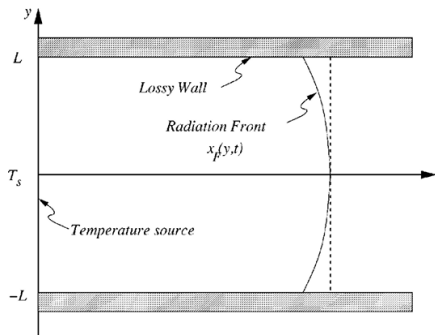
Goal

Build a 2D semi-analytical model to understand supersonic heat wave propagation and temperature distribution.

Meaning, Understanding the following equation:

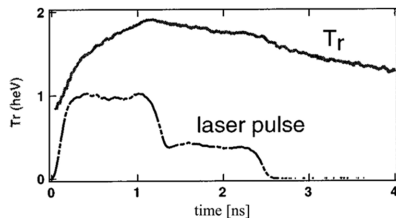
$$\frac{\partial T^4}{\partial t} + f \rho^{-\mu+1} \frac{\partial T^\beta}{\partial t} = \nabla \cdot (D(T) \nabla (a T^4))$$

using cylindrical coordinates and the relevant boundary conditions.

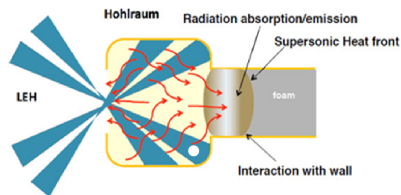


Marshak Waves Experiments

Using a Hohlraum as an x-ray generator, a supersonic heatwave propagates in the low-density foam.



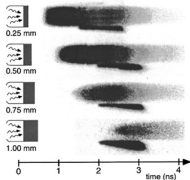
Back et al. POP, (2000)



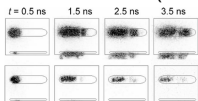
Cohen et al. PRR, (2020)

Laser $\Rightarrow$ hohlraum $\Rightarrow$ x-ray radiation $\Rightarrow$ supersonic heat wave propagates in the low-density foam

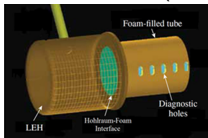
Marshak Waves Experimental Setup



Back et al. POP, (2000)

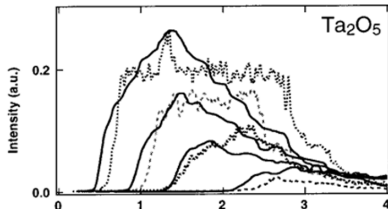


Rosen et al. ASS, (2006)

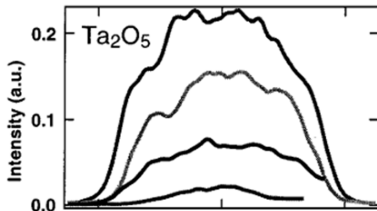


Cohen et al. PRR, (2020)

There are several ways to track the heat wave propagation in experiments:



Back et al. POP, (2000)



Experiments tested

The experiment	foam type	density (mg/cc)	temperature (eV)
Massen <i>et al.</i>	$C_{11}H_{16}Pb_{0.3852}$	80	100–150
Back <i>et al.</i>	SiO_2 , Ta_2O_5	10–50	85–190
Xu <i>et al.</i>	C_6H_{12} , $C_6H_{12}Cu_{0.394}$	50	160
Ji-Yan <i>et al.</i>	C_8H_8	160	175
Keiter <i>et al.</i>	$C_{15}H_{20}O_6$, $C_{15}H_{20}O_6Au_{0.172}$	45–70	200–210
Moore <i>et al.</i>	SiO_2 , C_8H_7Cl	90–120	310
Courtois <i>et al.</i>	SiO_2	19 – 29	170

1.5D Model

HR Model - Governing Equations

Hammer and Rosen solved the diffusion equation with the specific energy and opacity power laws:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT^\alpha \rho^{-\lambda}$$

where $f, \mu, g, \lambda, \alpha, \beta$ are material-dependent parameters:

Assuming $\rho f T^\beta \rho^{-\mu+1} \gg a T^4$ (true until around 3keV), HR solved:

$$f \rho^{-\mu+1} \frac{\partial T^\beta}{\partial t} = \frac{\partial}{\partial x} \cdot \left(\frac{c g T^\alpha \rho^{-\lambda-1}}{3} \frac{\partial}{\partial x} (a T^4) \right)$$

Using perturbation theory, HR derived key physical quantities.

Rosen et al. POP, (2003)

HR Model - Heat Front Solution

The heat front position $x_F(t)$ is given by:

$$x_F^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt'$$

where:

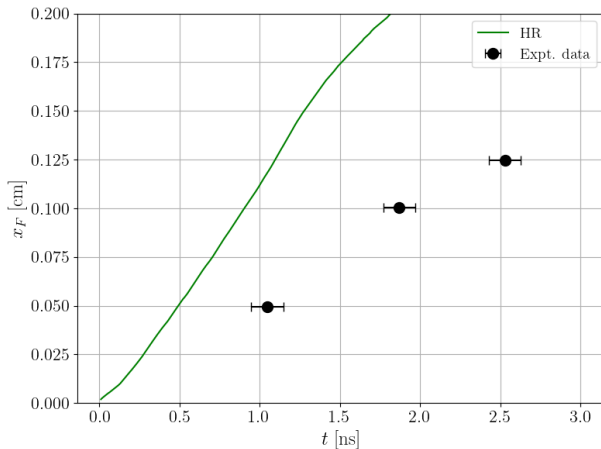
$$C = \frac{16g\sigma_{SB}}{3f(4 + \alpha)} \rho^{2-\mu-\lambda}; \quad \varepsilon = \frac{\beta}{4 + \alpha}$$

$$H(t) = T_s^{4+\alpha}(t)$$

where T_s is the **surface temperature** at $x = 0$.

HR 1D Model Limitations

Naive HR misses the experimental results of Back et al. by a factor of 2!



Conclusion: assuming $T_s = T_D$ overestimates the wave speed.

Marshak Boundary Condition: $T_D \neq T_s$

The Marshak boundary condition states that

$$\sigma_{SB}T_D^4(t) = \int_0^1 I(\mu)\mu d\mu = \frac{cE(0,t)}{4} + \frac{F(0,t)}{2}$$

where $F(0,t)$ is the net flux entering the foam.

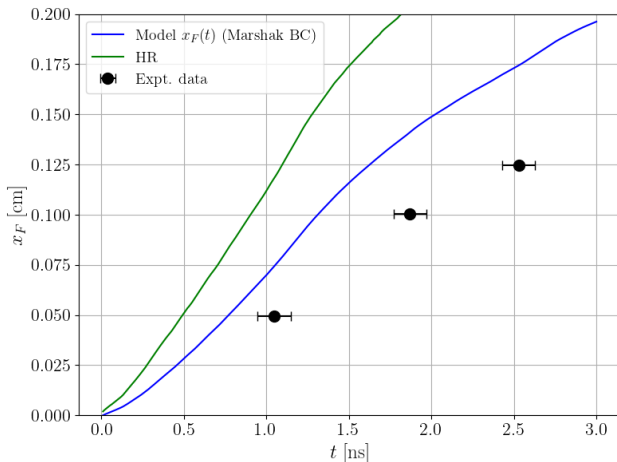
Leading to,

$$\sigma_{SB}T_D^4(t) = \sigma_{SB}T_s^4(t) + \frac{F(0,t)}{2}$$

Therefore, the surface temperature T_s is **always lower** than the drive temperature T_D !

Marshak Boundary Condition: $T_D \neq T_s$

Applying the Marshak boundary condition slows the propagation front significantly:



1.5D Model

2D Effects

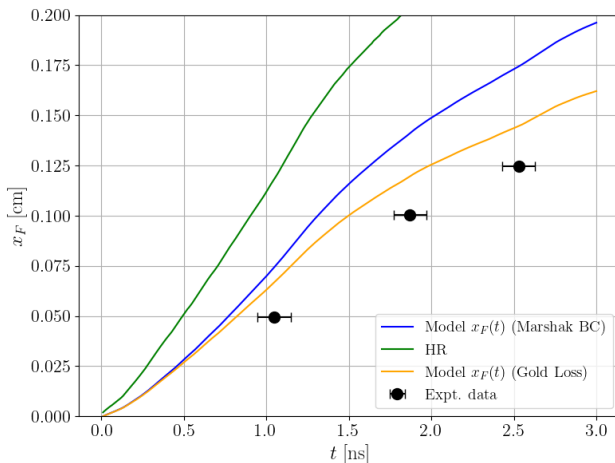
- Energy is lost from the foam into the tube walls.
- Can be modeled for different wall materials: **Gold**, **Copper**, **Beryllium**, and **Vacuum** (without external coating material).
- From Ramot's fits, we can take the subsonic energy of gold:

$$E_{\text{gold}}(t) = 0.59 T_s^{3.35} [t - t_0(x)]^{0.59} \quad [\text{hJ}/\text{mm}^2]$$

where $t_0(x)$ is the front arrival time at position x .

Energy Wall Loss

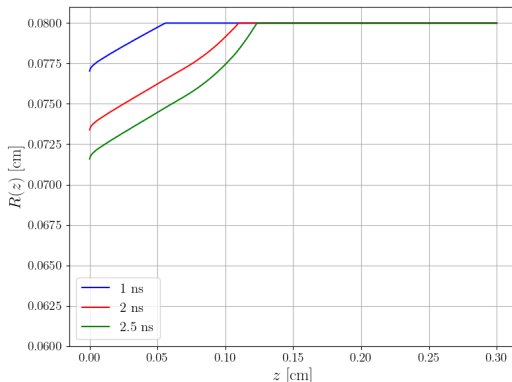
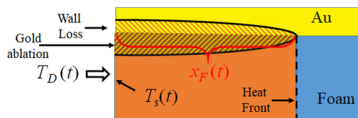
Including wall loss reduces the front velocity:



Wall Ablation

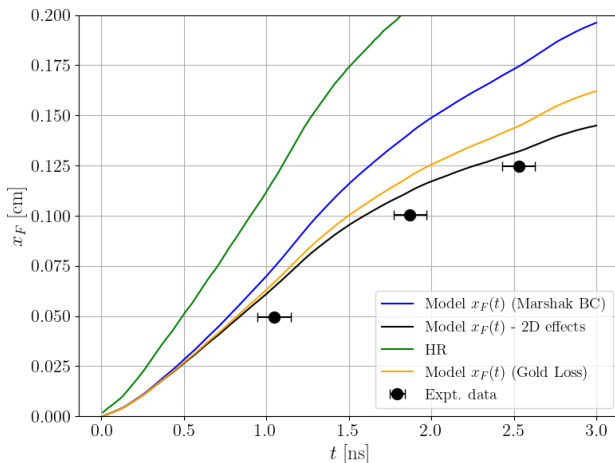
- High-Z wall material (e.g. Gold) absorbs energy and ablates inward into the foam.
- **Changing Foam Radius:** The inlet and tube radius shrink, reducing the heat wave cross-section.
- **Changing Foam Density:** The ablated wall material compresses and densifies the adjacent foam:

$$\rho(t) = \rho_0 \frac{V_0}{V(t)}$$



Wall Ablation

Inward wall ablation compressively densifies the foam, which significantly slows down propagation:



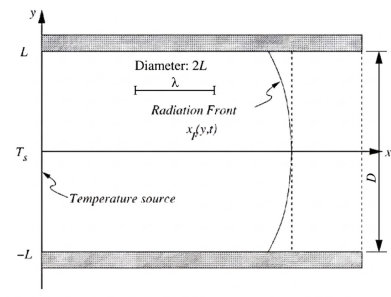
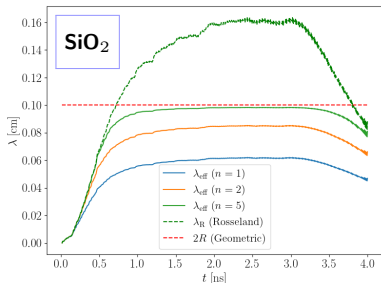
Effective Mean Free Path

When $\lambda_R \geq 2R$, wall collisions control the transport instead of diffusion.
Can be modeled as (*Courtois et al. POP, (2024)*):

$$\frac{1}{\lambda_{\text{eff}}} = \frac{1}{\lambda_R} + \frac{1}{2R}, \quad \lambda_R \text{ is the Ross Mean Free Path}$$

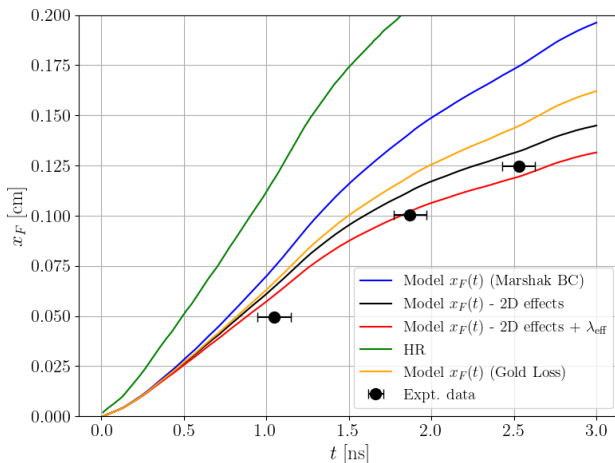
A smoother transition is defined using the generalized power mean:

$$\lambda_{\text{eff}} = \left((2R)^{-n} + \lambda_R^{-n} \right)^{-1/n}$$

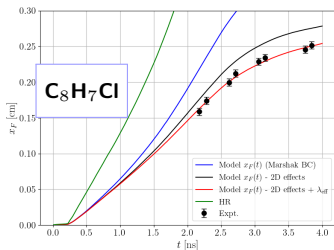


Effective Mean Free Path

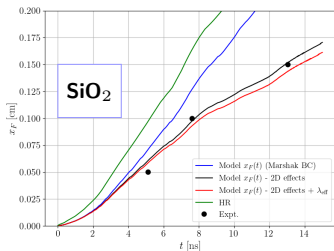
Including the effective mean free path correction matches the model to the experimental data:



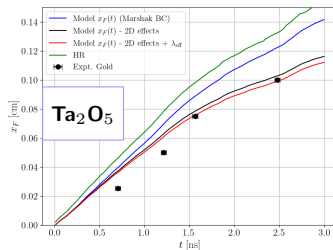
Comparison to many experiments



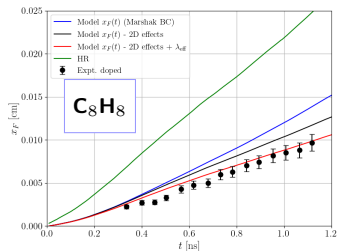
Moore et al. JQSRT, (2015)



Back et al. PRL, (2000)



Back et al. POP, (2000)



Ji-Yan et al. CPB, (2010)

Flux out of the model

Using the Henyey-like profile:

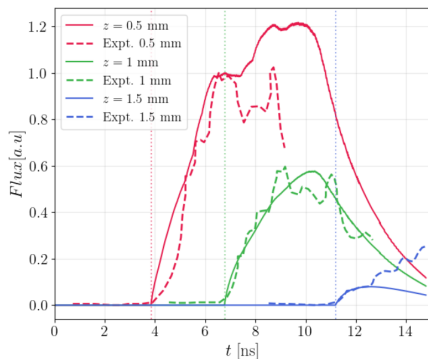
$$T(z, t) = T_S(t) \left[1 - \frac{z}{z_F(t)} \right]^{\frac{1}{4+\alpha-\beta}}$$

behind the heat front ($z < z_F(t)$),
and $T = 0$ ahead of it.

The radiation flux at each detector
position is then computed as:

$$\Phi(t) = \sigma_{\text{SB}} T^4(z, t)$$

where the flux is computed in several
different spatial locations.



2D Model

Full 2D Model

Modeling the Front Curvature

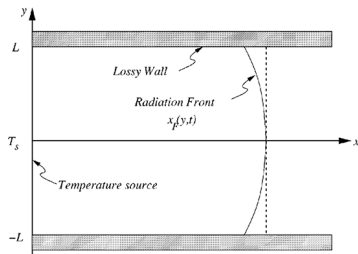
Understanding the front curvature

Hurricane & Hammer (2006) solved the Laplace equation (assumed constant opacity and neglected e):

$$\underbrace{\frac{\partial e}{\partial t}}_{\rightarrow 0} = \frac{4}{3} \nabla \cdot \left(\frac{1}{\rho \kappa} \nabla (T^4) \right)$$

and got the 2D front in **Cartesian coordinates** (where $\varepsilon = \frac{3}{4} \rho \kappa L (1 - \alpha)$):

$$x_F(y, t) = \frac{L}{\varepsilon} \cosh^{-1} \left(1 + \frac{D \varepsilon t}{L^2} \right) \cos \left(\frac{\sqrt{\varepsilon}}{L} y \right) + \text{HOT}$$



Cylinder Coordinates

The model was re-developed in **cylindrical coordinates** (*Courtois et al. POP, (2022)*):

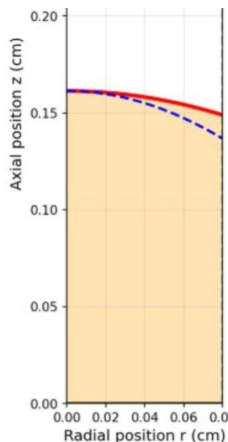
$$x_F(r, t) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{D\kappa_0^2 t}{2} \right) J_0(\kappa_0 r) + HOT$$

Planar $\longleftrightarrow$ **Cylindrical Analogy:**

$\cos(ky) \longleftrightarrow J_0(\kappa r)$ For small ε :

$$\kappa_0 \approx \frac{\sqrt{2\varepsilon}}{R}, \quad \varepsilon = \frac{3}{4} \rho \kappa R (1 - \alpha)$$

where α is the wall albedo.



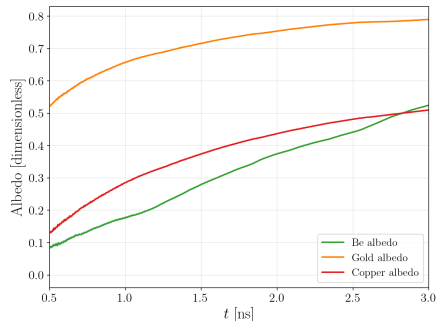
Applying on our model: Dynamic Albedo

In previous works, the wall albedo was treated as a constant fitted parameter. We update it dynamically in time:

$$\begin{aligned}\alpha(t) &= \frac{F_{\text{out, wall}}}{F_{\text{in, wall}}} \\ &= \frac{\sigma_{\text{SB}} T_{s, \text{interface}}^4 - \frac{dE_{\text{wall}}}{dt} / 2}{\sigma_{\text{SB}} T_{s, \text{interface}}^4 + \frac{dE_{\text{wall}}}{dt} / 2}\end{aligned}$$

where the curvature goes as

$J_0 \left(\frac{\sqrt{2\varepsilon(t)}}{R} r \right)$, and $\varepsilon(t)$ scales with $(1 - \alpha(t))$.



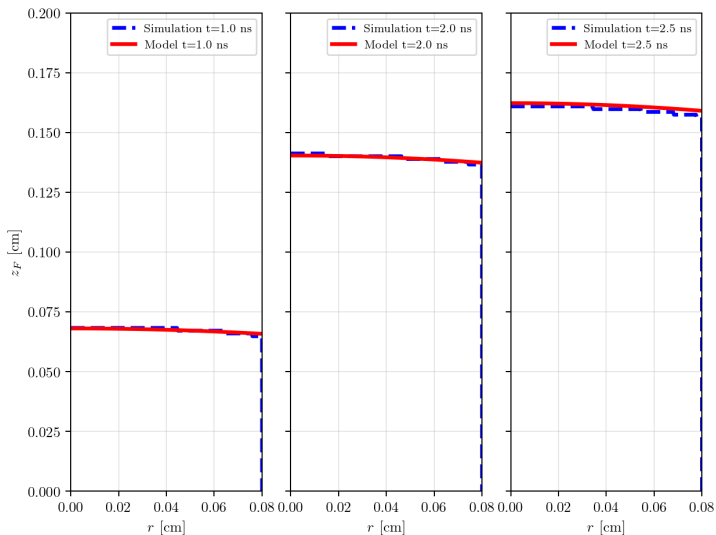
The 2D wave front

$$\boldsymbol{x}_F(\boldsymbol{r}, t) = \boldsymbol{x}_{1.5\text{D}}(t) \cdot J_0(\kappa_0(t)\boldsymbol{r})$$

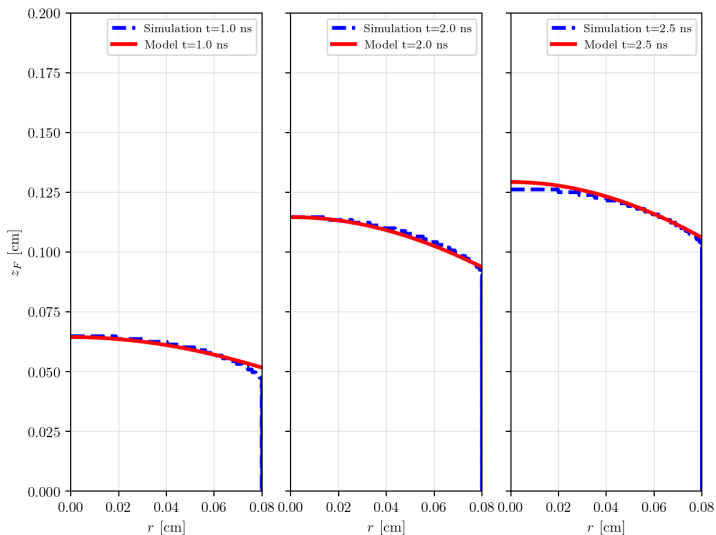
Full 2D Model

**Comparing to 2D Diffusion
Simulation without hydrodynamics**

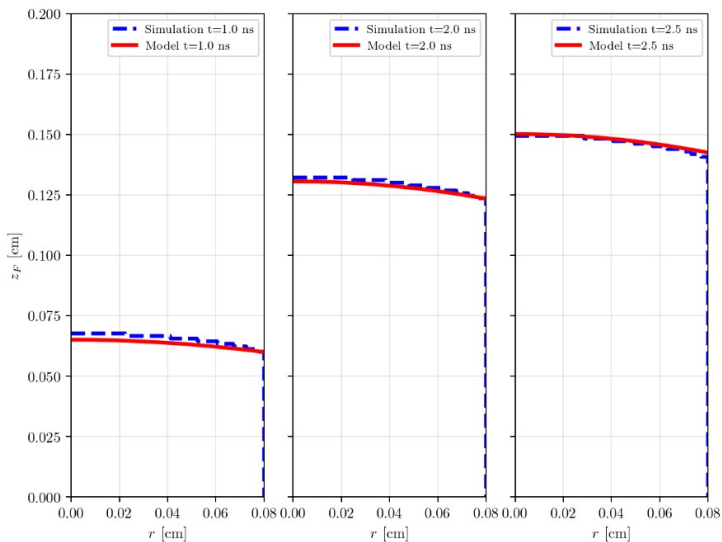
Gold coating - SiO₂

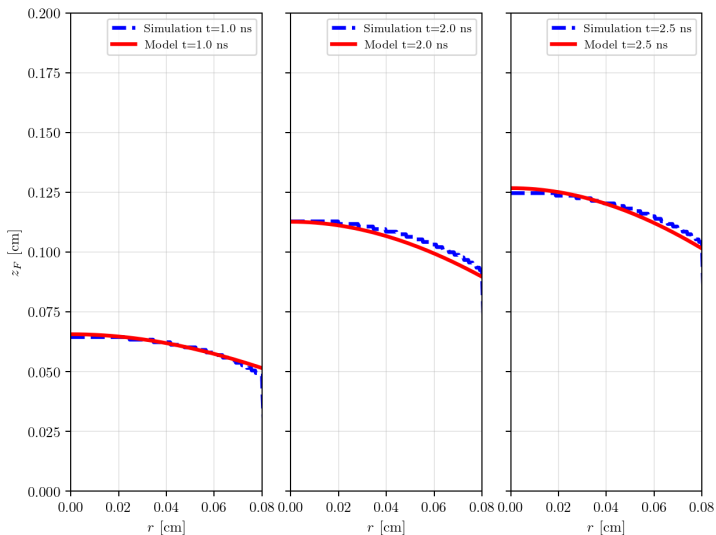


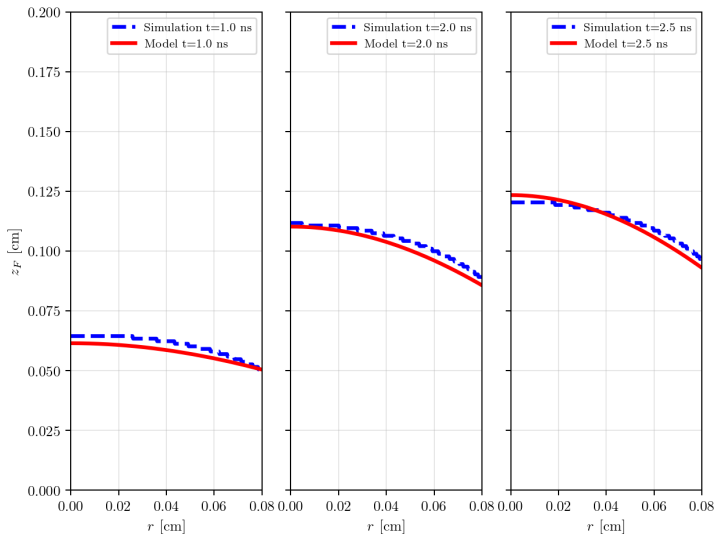
100 times optically thinner Gold coating - SiO₂



Copper - SiO₂







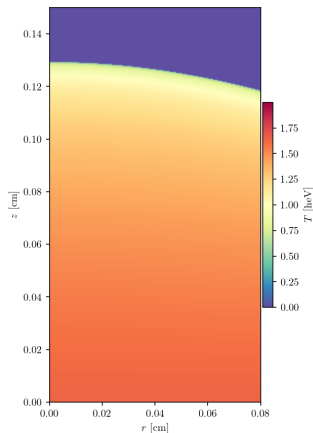
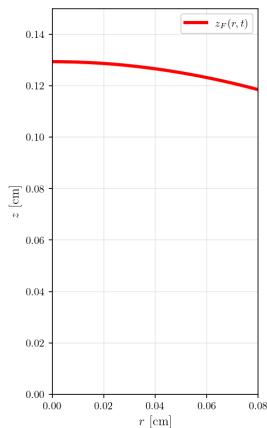
Full 2D Model

A Full 2D Temperature Distribution

A full 2D Temperature distribution

One can use the 2D Henyey-like profile:

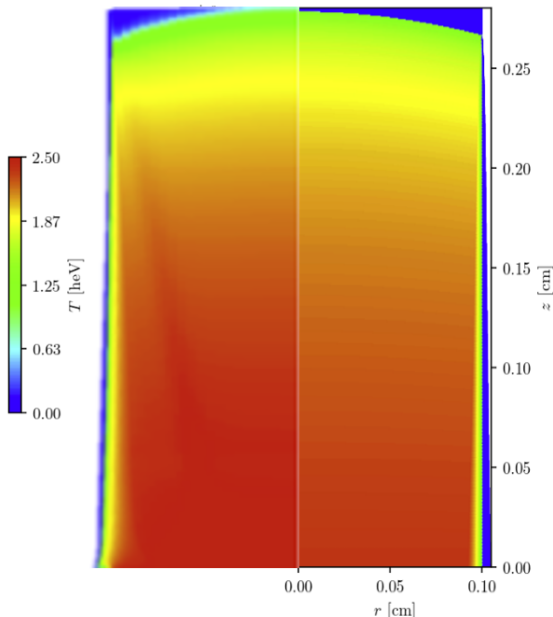
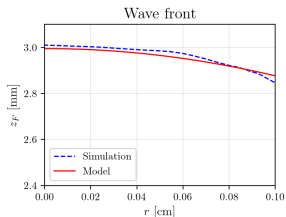
$$T(z, r, t) \approx T_S(t) \left[1 - \frac{z}{z_F(r, t)} \right]^{\frac{1}{4+\alpha-\beta}}$$



2D Temperature Distribution — Moore (C_8H_7Cl)

Side-by-side comparison of the 2D temperature distribution at $t = 3.72$ ns:

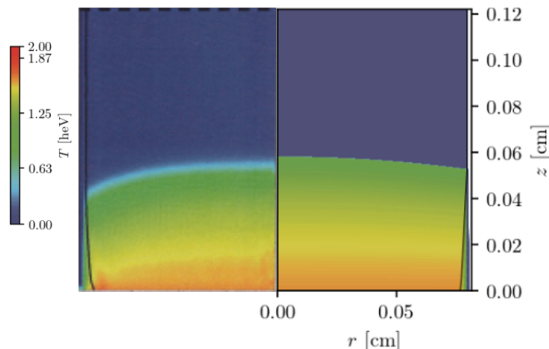
- **Left:** Diffusion simulation with hydrodynamics taken from *Moore et al. JQSRT, (2015)*.
- **Right:** Our full 2D model.



2D Temp. Distribution — Back SiO₂ (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

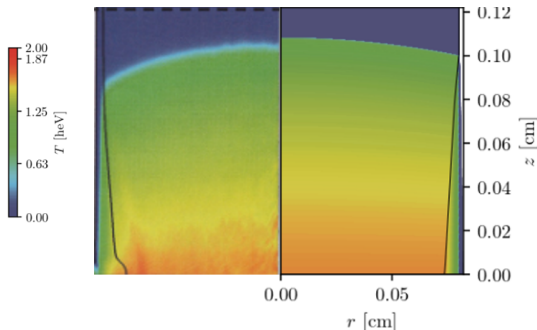
- **Left:** Implicit Monte Carlo (IMC) simulation from *Cohen et al. PRR, (2020)*.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temp. Distribution — Back SiO₂ (2.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.0$ ns:

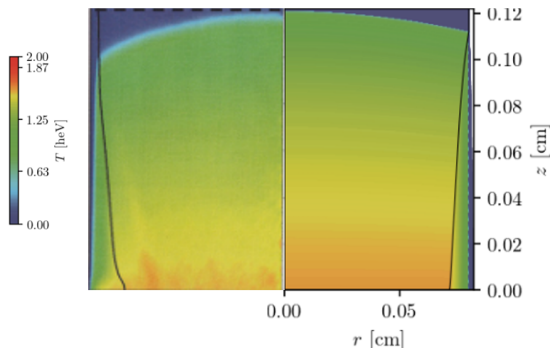
- **Left:** Implicit Monte Carlo (IMC) simulation from *Cohen et al. PRR, (2020)*.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temp. Distribution — Back SiO₂ (2.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.5$ ns:

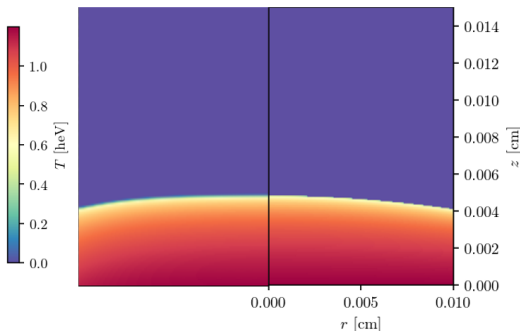
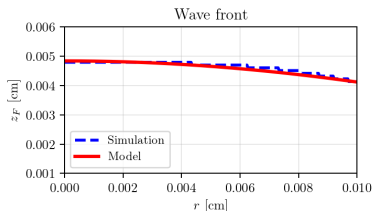
- **Left:** Implicit Monte Carlo (IMC) simulation from *Cohen et al. PRR, (2020)*.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temp. Distribution — Ji-Yan (0.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 0.5$ ns:

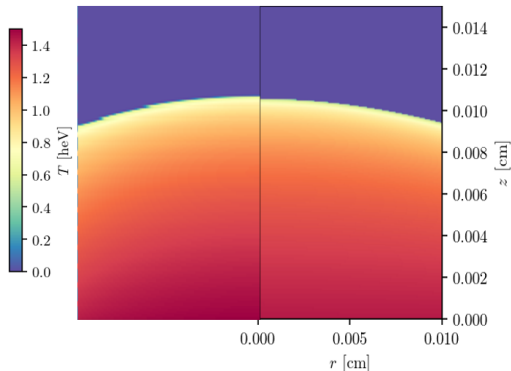
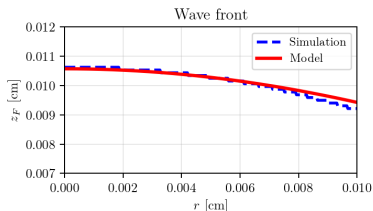
- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



2D Temp. Distribution — Ji-Yan (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

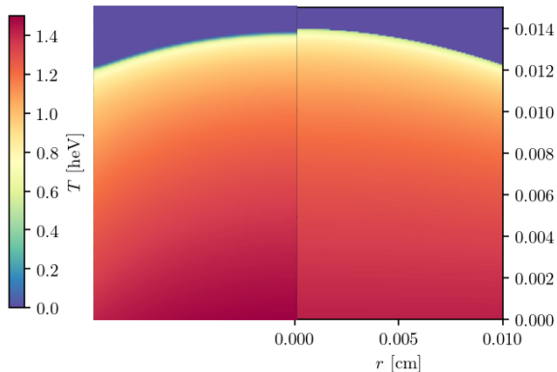
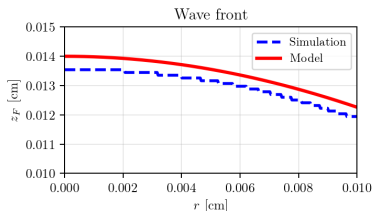
- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



2D Temp. Distribution — Ji-Yan (1.3 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.3$ ns:

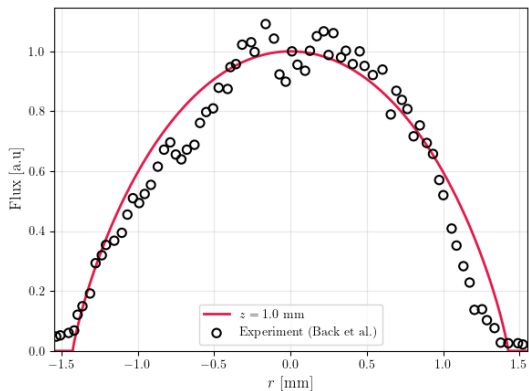
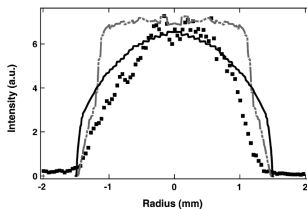
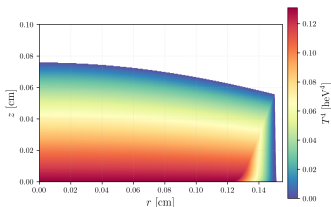
- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



Full 2D Model

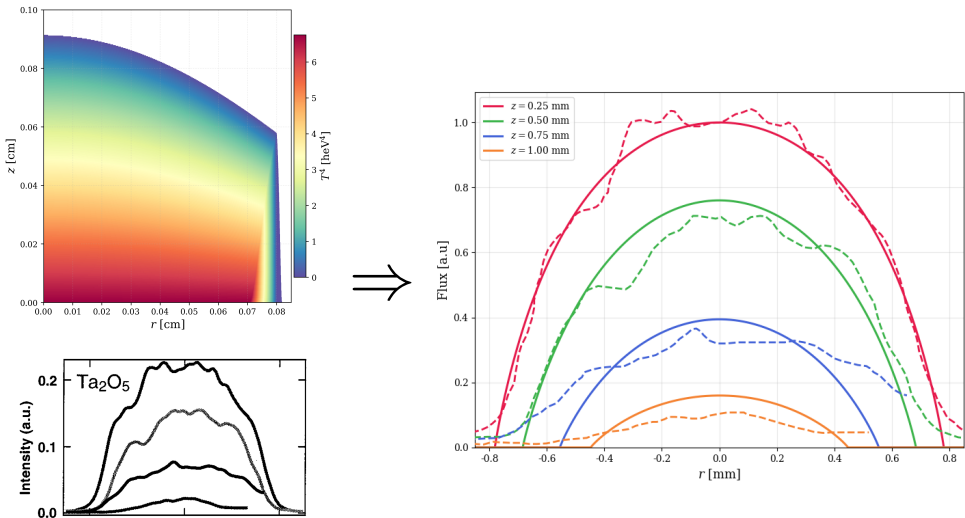
Comparison with Experimental Data

Flux Curvature Comparison - Back SiO₂ PRL



Back et al. PRL, (2000)

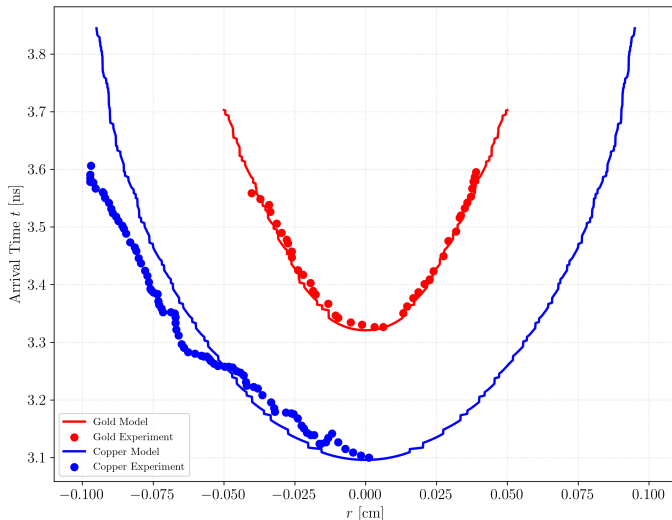
Flux Curvature Comparison - Back Ta2O5 POP



Back et al. POP, (2000)

Gold vs. Copper Coating Comparison

French experiment *Courtois et al. POP, (2024)* which tests the arrival time of the front as a function of r for both gold and copper coats:



- **Successful 2D Modeling:**
 - We have successfully modeled 2D supersonic Marshak waves.
 - Our unified model captures front curvature and temperature distribution under the diffusion approximation.
- **Deep Physical Insight:**
 - We can now understand the underlying physics of radiation transport and energy exchange in a very good, intuitive way.

Thank You!

Questions?